

Business Requirements Document (BRD) & Operations Strategy

Project: Hospital Workflow Optimization & Centralized EMR System Integration

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Domain: Healthcare Operations & Systems Engineering

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1. Executive Summary & Problem Definition

The hospital currently faces critical operational bottlenecks across patient intake, clinical consultations, and internal prescription routing:

- **Severe Waiting Delays:** Average patient waiting time from arrival to physical consultation exceeds 90 minutes.
- **Vulnerable Record Systems:** Legacy paper records require duplicate data entry, cause physical file loss, and impede patient history lookup[cite: 14, 15].
- **Communication Silos:** Lack of digital interconnectivity between admissions, consulting doctors, the pathology lab, and the pharmacy leads to prescription fulfillment errors and billing bottlenecks[cite: 14, 15].

2. Project Scope & Objectives

Primary Objectives:

- Reduce patient check-in and waiting times by 40% within 6 months of implementation[cite: 14, 15].
- Deploy a centralized, cloud-hosted Electronic Medical Record (EMR) system integrated with pharmacy and laboratory modules[cite: 14, 15].
- Replace manual paper prescription transfers with digital physician order entry (e-Prescribing) to eliminate transcript errors[cite: 14, 15].

Project Boundaries:

- **In-Scope:** Digital patient pre-registration/intake kiosk, centralized doctor EMR lookup, automated e-prescribing queue for the dispensary pharmacy, and billing synchronization[cite: 14, 15].
- **Out-of-Scope:** Clinical equipment procurement, nurse payroll automation, and structural physical facility remodeling.

3. Stakeholder Analysis Matrix

Stakeholder Group	Project Role	Influence	Interest	Key Requirements & Expectations
Hospital Administration / Board	Project Sponsor & Budget Approval	High	High	ROI, regulatory compliance, increased patient throughput.
Attending Physicians / Doctors	Primary Clinical End-Users	High	High	Sub-second patient medical history lookup; minimal screen time during patient rounds.
Nursing & Triage Staff	Operational Intake Staff	Medium	High	Simplified vital entry screens; elimination of paper clipboards.
Pharmacy & Lab Heads	Fulfillment Operations	Medium	High	Legible e-prescriptions; automated real-time inventory decrementing
Hospital IT Operations	Infrastructure & Security	High	Medium	99.9% uptime SLA, role-based access control, HIPAA/data privacy compliance.
Patients	Service Beneficiaries	Low	High	Minimal wait times, transparent digital billing, secure health records.

4. Requirements Specification (Functional & Non-Functional)

4.1 Functional Requirements (FR)

- FR-01 (Digital Intake): The system shall provide a self-service check-in interface via mobile QR code and on-site tablet kiosks to capture patient details and reason for visit.
- FR-02 (Centralized EMR Access): The system shall grant authenticated clinicians instant, single-click access to consolidated patient records, including past diagnoses, allergy alerts, and laboratory history.
- FR-03 (Automated e-Prescriptions): The system shall allow physicians to create and digitally transmit electronic prescriptions directly into the hospital pharmacy fulfillment queue.
- FR-04 (Role-Based Access Control): The system shall enforce strict data access tiers (e.g., billing administrators cannot view physician clinical progress notes).

4.2 Non-Functional Requirements (NFR)

- NFR-01 (Data Security & Compliance): All patient identifiable data must be encrypted at rest (AES-256) and in transit (TLS 1.3), adhering strictly to healthcare data privacy and HIPAA standards.
- NFR-02 (Performance & Latency): Patient record queries must return search results in under 1.5 seconds under peak simultaneous hospital loads.
- NFR-03 (Availability & Backups): System infrastructure must ensure 99.9% uptime with automated incremental backups scheduled every 4 hours to prevent data loss.

5. Gap Analysis & Solution Prioritization

5.1 Operational Gap Analysis

Operational Area	Current State (As-Is)	Desired State (To-Be)	Identified Gap	Action Plan
Patient Registration	Manual paper forms, physical queue tokens.	Digital self-service check-in kiosk & mobile QR check-in.	No digital front-end intake capture.	Deploy touch-screen kiosks with automated token routing.
Medical Records	Physical paper folders transported manually between floors.	Centralized cloud-based EMR with instant lookup.	Decentralized, vulnerable paper records.	Deploy relational database with unified patient ID indexing.
Pharmacy Routing	Handwritten paper prescriptions carried by patients.	Direct electronic push to pharmacy POS terminal.	Manual hand-off latency and handwriting error risks.	Integrate physician order entry with dispensary dashboard.

5.2 MoSCoW Prioritization Framework

- Must Have (Core Scope): Centralized EMR database, role-based clinician/staff login, digital e-prescribing module, and end-to-end data encryption.
 - Should Have (High Value): Automated SMS appointment alerts and automated stock deduction in the pharmacy inventory database upon dispensation.
 - Could Have (Desirable): Dedicated mobile patient portal allowing self-download of completed lab reports.
 - Won't Have (Phase 2): AI-driven predictive clinical diagnosis and automated bed allocation algorithms.
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